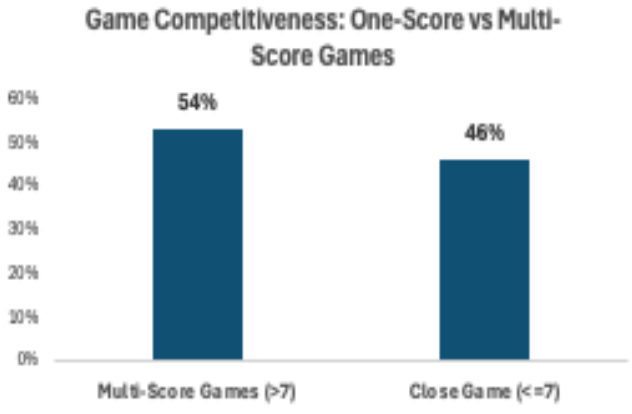
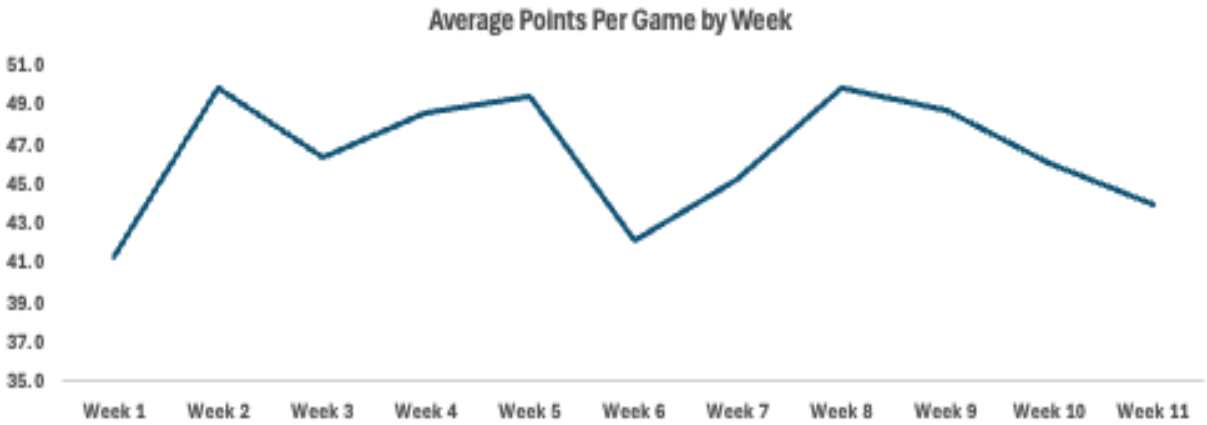
NFL Season Overview

Total Games	AVG Points/Game	Total Points	Home Win %	AVG Margin
213	45.2	9624	52.1%	11.3

Home Average Point	Away Average Points
23.7	21.5







Highest Scoring Games

Date	AwayTeam	HomeTeam	Total Points
11/2/25	Bears	Bengals	89
9/7/25	Ravens	Bills	81
9/28/25	Packers	Cowboys	80

Lowest Scoring Games

Date	AwayTeam	HomeTeam	Total Points
11/6/25	Raiders	Broncos	17
11/10/25	Eagles	Packers	17
10/19/25	Panthers	Jets	19

NFL ANALYTICS PROJECT – DASHBOARD SQL

Purpose: This project analyzes NFL game-level data to identify scoring patterns, team performance, and overall competitiveness during the season. Using SQL, I aggregated and transformed raw game data into league-wide KPIs, team-level metrics, and weekly trends. The results were visualized in an Excel dashboard designed to communicate insights at a glance clearly.

Key areas of analysis include league scoring averages, home versus away performance, top-scoring teams, game competitiveness (one-score versus multi-score games), and scoring trends over time. All queries were limited to final games to ensure accuracy and consistency across metrics.

The goal of this project was to demonstrate end-to-end analytical workflow skills — from data validation and SQL aggregation to dashboard design and storytelling.

```
============================================================
  SECTION 1: DATASET SANITY CHECKS
============================================================


-- Total rows in dataset
SELECT COUNT(*) AS total_rows
FROM nfl_stats;

-- Games by status (FINAL, UPCOMING, BYE)
SELECT
  GameStatus,
  COUNT(*) AS games
FROM nfl_stats
GROUP BY GameStatus;


============================================================
  SECTION 2: LEAGUE-WIDE KPI SUMMARY (1 ROW)
  Use for KPI cards in Excel
============================================================

SELECT
  COUNT(*) AS total_games,
  AVG(HomeScore + AwayScore) AS avg_points_per_game,
  SUM(HomeScore + AwayScore) AS total_points_scored,
  AVG(HomeWin) AS home_win_pct,
```

```
  AVG(ABS(HomeScore - AwayScore)) AS avg_margin_of_victory
FROM nfl_stats
WHERE GameStatus = 'FINAL';


============================================================
   SECTION 3: HOME VS AWAY PERFORMANCE
============================================================

-- Average points scored by home vs away teams
SELECT
  AVG(HomeScore) AS avg_home_points,
  AVG(AwayScore) AS avg_away_points
FROM nfl_stats
WHERE GameStatus = 'FINAL';

-- Overall home win percentage
SELECT
  AVG(HomeWin) AS home_win_percentage
FROM nfl_stats
WHERE GameStatus = 'FINAL';


============================================================
   SECTION 4: TRUE TEAM PERFORMANCE (HOME + AWAY COMBINED)
   Powers team leaderboards and comparisons
============================================================

SELECT
  team,
  COUNT(*) AS games_played,
  AVG(points_scored) AS avg_points_scored,
  AVG(win) AS win_pct
FROM (
  SELECT
    HomeTeam AS team,
    HomeScore AS points_scored,
    HomeWin AS win
  FROM nfl_stats
  WHERE GameStatus = 'FINAL'

  UNION ALL

  SELECT
```

```
    AwayTeam AS team,
    AwayScore AS points_scored,
    CASE WHEN HomeWin = 0 THEN 1 ELSE 0 END AS win
  FROM nfl_stats
  WHERE GameStatus = 'FINAL'
) t
GROUP BY team
ORDER BY avg_points_scored DESC;


============================================================
   SECTION 5: WEEKLY SCORING TRENDS
   Use for line charts
============================================================

SELECT
  Week,
  AVG(HomeScore + AwayScore) AS avg_points_per_game
FROM nfl_stats
WHERE GameStatus = 'FINAL'
GROUP BY Week
ORDER BY Week;


============================================================
   SECTION 6: GAME TYPE BREAKDOWN
   Close games vs blowouts
============================================================

SELECT
  CASE
    WHEN ABS(HomeScore - AwayScore) <= 7 THEN 'Close Game (≤7)'
    ELSE 'Blowout (>7)'
  END AS game_type,
  COUNT(*) AS games,
  AVG(HomeScore + AwayScore) AS avg_points
FROM nfl_stats
WHERE GameStatus = 'FINAL'
GROUP BY game_type;


============================================================
   SECTION 7: HIGHEST SCORING GAMES
   Storytelling/highlight table
```

```
============================================================

SELECT
  Date,
  AwayTeam,
  HomeTeam,
  (HomeScore + AwayScore) AS total_points
FROM nfl_stats
WHERE GameStatus = 'FINAL'
ORDER BY total_points DESC
LIMIT 10;


============================================================
```

SECTION 8: LOWEST SCORING GAMES

```
============================================================

SELECT
  Date,
  AwayTeam,
  HomeTeam,
  (HomeScore + AwayScore) AS total_points
FROM nfl_stats
WHERE GameStatus = 'FINAL'
ORDER BY total_points ASC
LIMIT 10;
```